Computational Statistics and Simulation

Worksheet 2: Monte Carlo Methods in Statistical Inference and Resampling Methods — *bootstrap* and *Jackknife*

Academic year 2025/26

1. Obtain estimates of $p = P(X > 2)$, with $X \sim C(0, 1)$, i.e.,

$$f(x) = \frac{1}{\pi(1 + x^2)}, \qquad -\infty < x < \infty,$$

using

   (a) the standard Cauchy distribution;

   (b) the Uniform distribution on $(0, 2)$;

   (c) it is known that $p = 0.14758$. Compare the results obtained above and their variances.

2. The aim is to compute the value of the integral

$$I = \int_0^\infty \log(x)\, x^5 e^{-3x}\, dx.$$

Using the Monte Carlo integration method, obtain estimates by simulating from

   (a) the distribution $X \sim Exp(3)$;

   (b) the distribution $X \sim Ga(6, 3)$;

   (c) also use the replication method to estimate I with $X \sim Ga(6, 3)$;

   (d) it is known that $I = 0.1000009$. Compare the results obtained above with the exact value and with the estimate of the integral (and its error) in the case of the replication method.

3. The aim is to estimate the integral

$$\int_0^1 \frac{e^{-x}}{1 + x^2}\, dx$$

using the importance sampling technique. Consider the following candidate importance functions:

$$f_1(x) = 1,\ 0 < x < 1; \qquad f_2(x) = e^{-x},\ 0 < x < \infty; \qquad f_3(x) = \frac{1}{\pi(1 + x^2)},\ -\infty < x < \infty;$$

$$f_4(x) = \frac{e^{-x}}{1 - e^{-1}},\ 0 < x < 1; \qquad f_5(x) = \frac{4}{\pi(1 + x^2)},\ 0 < x < 1.$$

Determine the associated errors and analyse which ones reduce the variance of the integral estimate the most.

4. Let $X \sim N(1, \sigma^2)$. It is known that

$$\left] \frac{(n-1)S_c^2}{\chi^2_{1-\alpha/2,\, n-1}}\ ,\ \frac{(n-1)S_c^2}{\chi^2_{\alpha/2,\, n-1}} \right[$$

is a $100(1 - \alpha)\%$ confidence interval for σ^2. Using the Monte Carlo method, determine a confidence interval for σ^2.

5. Consider a time series defined by $X_t = aX_{t-1} + Y_t$, where $Y \sim N(0, \sigma^2)$. The least squares estimator of a is given by

$$T = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}.$$

Assuming the model with parameters $a = 0.3$ and $\sigma^2 = 1$, a sample of 100 observations, and using the replication method (use 1000 replications):

   (a) obtain an estimate of the parameter a;

   (b) compute the standard deviation, mean squared error and bias associated with the estimate obtained; draw the boxplot of the sample of biases;

   (c) determine a confidence interval for a.

6. Suppose that one wants to

   (a) test $H_0 : \sigma^2 = 4$ vs $H_1 : \sigma^2 > 4$, at significance level $\alpha = 0.05$. Using 100000 replications, obtain estimates of the significance level of the test, considering samples of size 100 from normal populations with unit mean, taking $\sigma^2 = 4$, $\sigma^2 = 10$ and $\sigma^2 = 1$. Comment on the results.

   (b) determine a confidence interval for σ^2, using the Monte Carlo method and data drawn from $N(1, 4)$.

7. The data in the file `"tabaco.txt"` come from a study on the relationship between tobacco and lung cancer. Twenty-five occupational groups were studied. The explanatory variable x is the number of cigarettes smoked per day by male individuals in each occupational group. The response variable y is the standardised mortality rate due to deaths from lung cancer.

   (a) Plot the data and obtain the empirical correlation coefficient.

   (b) Generate 1000 bootstrap samples of the data, obtain the corresponding correlation coefficient and compute an estimate of the standard error of the correlation coefficient.

8. Consider the data on the duration of the eruptions of the *Old Faithful* geyser — available in the R library `MASS` as `faithful$eruptions`. The aim is to determine a confidence interval for the mean of the population underlying the data. Using the estimator $\bar{X}$, determine that confidence interval by the bootstrap resampling method.

9. For a sample $X_1, \ldots, X_n$ i.i.d. as X, consider the estimator $\hat{\theta}_n = \bar{X}$ of the mean of X. We then know that $se(\hat{\theta}_n) = \sqrt{\mathrm{Var}(X)/n}$. Show that

   (a) $\hat{\theta}_{(j)} = \dfrac{n\bar{X} - X_j}{n-1}$;

   (b) $\bar{\hat{\theta}}_{(\cdot)} = \bar{X}$;

   (c) $\widehat{se}_{jack}(\hat{\theta}_n) = \sqrt{S_n^2/n}$, with $S_n^2 = \dfrac{1}{n-1}\sum_{j=1}^{n}(X_j - \bar{X})^2$.

10. Solve exercise 7 using the Jackknife resampling method to estimate the correlation and compute an estimate of its bias.